

```
In [ ]: #How to import image
```

```
In [1]: import numpy as np #array
```

```
In [2]: import matplotlib.pyplot as plt #visualization
```

```
In [3]: from PIL import Image #python image library
```

```
In [6]: nature_image = Image.open(r'C:\Users\SID Balurkar\Desktop\Full Stack\Classwork\pic.jpg')  
nature_image
```

Out[6]:



```
In [7]: my_image = Image.open(r'C:\Users\SID Balurkar\Desktop\Photos\photo.jpg')  
my_image
```


Out[7]:



```
In [8]: print(type(nature_image))  
print(type(my_image))
```

```
<class 'PIL.WebPImagePlugin.WebPImageFile'>  
<class 'PIL.JpegImagePlugin.JpegImageFile'>
```

```
In [ ]: #Convert to array
```

```
In [9]: nature_arr = np.asarray(nature_image)  
nature_arr
```

```
Out[9]: array([[137, 178, 203],
               [137, 178, 203],
               [138, 180, 204],
               ...,
               [158, 188, 212],
               [158, 188, 212],
               [158, 188, 212]],

               [[137, 178, 203],
               [137, 178, 203],
               [138, 180, 204],
               ...,
               [158, 188, 212],
               [158, 188, 212],
               [158, 188, 212]],

               [[137, 178, 203],
               [137, 178, 203],
               [138, 180, 204],
               ...,
               [158, 188, 212],
               [158, 188, 212],
               [158, 188, 212]],

               ...,

               [[ 64,  68,  72],
               [ 68,  72,  75],
               [ 70,  74,  77],
               ...,
               [ 28,  83, 110],
               [ 28,  83, 110],
               [ 28,  84, 108]],

               [[ 90,  96, 100],
               [ 72,  78,  83],
               [ 71,  77,  82],
               ...,
               [ 36,  88, 116],
               [ 35,  87, 115],
               [ 35,  87, 115]],
```



```
[[ 90,  96, 100],  
 [ 72,  78,  83],  
 [ 71,  77,  82],  
 ...,  
 [ 36,  88, 116],  
 [ 35,  87, 115],  
 [ 35,  87, 115]]], dtype=uint8)
```

```
In [10]: plt.imshow(nature_arr)  
plt.show()
```



```
In [11]: nature_arr.shape
```

```
Out[11]: (234, 325, 3)
```

```
In [ ]: #Similarly for my image
```

```
In [12]: myimage_arr = np.asarray(my_image)
myimage_arr
```

```
Out[12]: array([[ 96,  87,  80],
                [ 92,  88,  79],
                [ 91,  87,  76],
                ...,
                [255, 255, 255],
                [255, 255, 255],
                [255, 255, 255]],

               [[ 97,  90,  82],
                [ 96,  89,  81],
                [ 92,  88,  79],
                ...,
                [255, 255, 255],
                [255, 255, 255],
                [255, 255, 255]],

               [[102,  95,  85],
                [ 98,  91,  83],
                [ 93,  89,  80],
                ...,
                [255, 255, 255],
                [255, 255, 255],
                [255, 255, 255]],

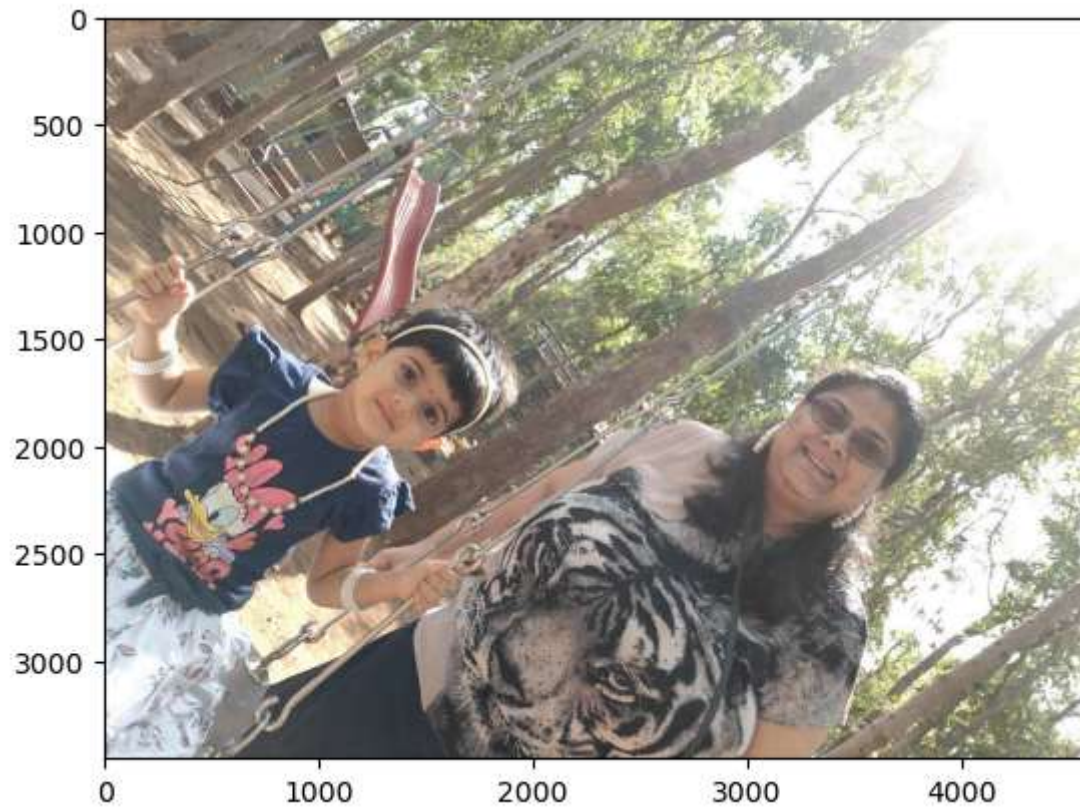
               ...,

               [[195, 198, 207],
                [211, 215, 224],
                [229, 233, 242],
                ...,
                [255, 255, 188],
                [255, 255, 192],
                [254, 255, 195]],

               [[220, 224, 233],
                [232, 236, 245],
                [245, 249, 255],
                ...,
                [255, 255, 186],
                [253, 255, 190],
                [253, 254, 194]],
```

```
[[225, 229, 238],  
 [234, 238, 247],  
 [243, 247, 255],  
 ...,  
 [255, 255, 188],  
 [253, 255, 190],  
 [253, 254, 194]]], dtype=uint8)
```

```
In [13]: plt.imshow(my_image)  
plt.show()
```



```
In [ ]:
```